TRYOLABS • INSTRUCTOR LIVE TROUBLESHOOTING & ATTENDEE Q&A PLAYBOOK

What Could Go Wrong During the Labs: 16 Common Attendee Questions & Exact Fixes

Keep this quick-reference playbook on your podium during the 4-hour Google Antigravity workshop (Telecom & Retail tracks). Every question below represents a realistic failure mode an attendee might run into during Labs 01–04, paired with the exact root cause, what to say to the attendee, and the copy-paste terminal or code fix.

[Environment & Setup (All Labs)] Q1: 'When I run ./labs/lab-01-.../install.sh, I get python3: command not found, No module named venv, or an old Python version error.'

Root Cause: Attendee laptop is missing Python 3.10+ on PATH or lacks python3-venv on Ubuntu/Debian/WSL.

Instructor Answer & Exact Fix:
1. Check if `uv` is installed (`uv --version`). If so, run `uv venv .venv --python 3.11 && source .venv/bin/activate && uv pip install -r requirements.txt`.
2. On macOS with Homebrew: `brew install python@3.11`.
3. On Windows: Run PowerShell script `.\labs\lab-01-...\install.ps1` instead of `.sh`.
4. Always ensure commands are run using `.venv/bin/python3` and `.venv/bin/pytest`.

[Environment & Setup (All Labs)] Q2: 'When I start the FastAPI dashboard (uvicorn src.main:app --port 8000), it crashes with [Errno 48] Address already in use (macOS/Linux) or WinError 10048 (Windows).'

Root Cause: Another process or a previous terminal tab is already listening on port 8000 (or port 8080 for `adk web`).

Instructor Answer & Exact Fix:
• On macOS/Linux, free the port immediately: `lsof -ti:8000 | xargs kill -9` (or for port 8080: `lsof -ti:8080 | xargs kill -9`).
• Alternatively, run on a different port: `.venv/bin/uvicorn src.main:app --reload --port 8005`.

[Environment & Setup (All Labs)] Q3: 'I accidentally corrupted my SQLite database (data/*.db) or deleted part of a service file while experimenting. How do I recover quickly?'

Root Cause: Attendee ran a destructive query or bad edit during exploration.

Instructor Answer & Exact Fix:
• Re-run the current lab's installer: `./labs/lab-0X-.../install.sh`.
• Explain that every `install.sh` script is **self-healing and idempotent**: it automatically restores the clean starter state for the current lab, applies verified solutions for all prior labs, and re-seeds the SQLite database (`python3 data/seed_telecom_data.py` or `python3 data/seed_retail_data.py`) in 2 seconds!

[Environment & Setup (All Labs)] Q4: 'I didn't finish Lab 1 in time and now the class is moving to Lab 2. Am I stuck?'

Root Cause: Normal pacing variance across attendees in a hands-on coding workshop.

Instructor Answer & Exact Fix:
• Reassure the attendee: 'Not at all! Just run `./labs/lab-02-rules-and-skills/install.sh`.'
• Every lab's `install.sh` automatically copies the reference solution from prior labs into `src/` before setting up the new lab. And if anyone wants to see the full solution to the current lab, they can run `./labs/lab-0X-.../apply_solution.sh`.

[Antigravity & Gemini Enterprise Auth] Q5: 'Antigravity gives a 403 PermissionDenied error or says cloudaicompanion.googleapis.com is not enabled for my account.'

Root Cause: The attendee's corporate Google identity is either logged into a personal Gmail profile or missing the IAM role binding on the workshop GCP project.

Instructor Answer & Exact Fix:
1. Verify the attendee is signed into Antigravity using their corporate email associated with the workshop.
2. In the Instructor GCP Console -> **IAM & Admin**, grant the attendee's email two roles: `Gemini for Google Cloud User` (`roles/cloudaicompanion.user`) and `Service Usage Consumer` (`roles/serviceusage.serviceUsageConsumer`).
3. Ensure `cloudaicompanion.googleapis.com` is enabled in **APIs & Services**.

[Lab 01 — Telecom Math & Logic] Q6 (Telecom): 'Antigravity updated bss_billing.py, but pytest still fails on the CDR charge amount by ~7.37% or a few cents!'

Root Cause: Two classic Telecom BSS pitfalls: (1) Dividing total bytes by decimal GB (`10**9 = 1,000,000,000`) instead of 3GPP binary GiB (`2**30 = 1024**3 = 1,073,741,824`); or (2) Rounding `volume_gib` to 2 decimals *before* multiplying by `base_rate` and `roaming_markup_multiplier`.

Instructor Answer & Exact Fix:
• Show the exact formula: `volume_gib = (bytes_uplink + bytes_downlink) / 1073741824.0`.
• Keep full floating-point precision during multiplication (`charge = volume_gib * rate_per_gib * multiplier`) and apply `round(charge, 2)` only on the final USD amount!
• Remind them to check `roaming_indicator != 'HOME'` (e.g., `'ROAMING_INTL'`) to apply `roaming_markup_multiplier`.

[Lab 01 — Telecom Math & Logic] Q7 (Telecom): 'Why is cell GNB-HOU-202 (URLLC slice) failing the SLA breach test when its PRB utilization is only 64% (below 85%)?'

Root Cause: Attendees (or a naive LLM prompt) assume a cell only breaches SLA when BOTH PRB > 85% AND latency > threshold.

Instructor Answer & Exact Fix:
• Explain the 5G domain reality: An Ultra-Reliable Low-Latency (`URLLC`, `SST=2`) slice for robotic surgery or smart grids has a strict `10ms` PDCP latency ceiling. Even if radio spectrum (`PRB`) is only 64% full, radio interference (`SINR`) can spike latency to `18ms`, violating the contract!
• In `src/services/ran_triage.py`, the condition must be `prb_utilization_pct > 85.0 or pdcp_latency_ms > sla_max_latency_ms`, sorted descending by `exposure_usd = sla_penalty_per_hour_usd * breach_duration_hours`.

[Lab 01 — Retail Math & Logic] Q8 (Retail): 'In margin_engine.py, I enforced a 15% floor using unit_cost_usd * 1.15, so why is pytest saying my minimum price is wrong?'

Root Cause: Classic confusion between **15% Markup** (`cost * 1.15`) and **15% Gross Margin** (`(price - cost) / price = 0.15`).

Instructor Answer & Exact Fix:
• Write both formulas on the screen—attendees love this clarification:
- **Markup** divides profit by cost: `price = cost * 1.15` (if cost is $85, 15% markup gives $97.75, which is only a 13.04% gross margin!).
- **Gross Margin** divides profit by net selling price: `(price - cost) / price >= 0.15`, which algebraically rearranges to `min_price = round(unit_cost_usd / (1.0 - 0.15), 2)` (if cost is $85, `$85 / 0.85 = $100.00`, giving an exact 15.0% gross margin).
• Also confirm `STACKABLE` promos are applied sequentially (`price = price * (1 - d1) * (1 - d2)`), never added (`d1 + d2`).

[Lab 01 — Retail Math & Logic] Q9 (Retail): 'Why does my GS1 GTIN-14 barcode validator reject valid 14-digit barcodes in inventory_atp.py?'

Root Cause: Reversing the 3-and-1 multiplier weights in the GS1 Modulo-10 algorithm.

Instructor Answer & Exact Fix:
• For a 14-digit `GTIN-14`, the first 13 digits (`digits[:13]`) are the payload and the 14th digit (`digits[13]`) is the check digit.
• Moving from left to right across the first 13 digits (index `0` to `12`), even indices (`0, 2, 4, ... 12`) are multiplied by **`3`**, and odd indices (`1, 3, 5, ... 11`) are multiplied by **`1`**. Then `check_digit = (10 - (total % 10)) % 10`.

[Lab 02 — Rules & Skills] Q10: 'I created SKILL.md inside .agents/skills/, why doesn't Antigravity discover or activate my skill when I prompt it?'

Root Cause: Progressive disclosure discovery relies strictly on the YAML frontmatter at the top of `SKILL.md` and the exact directory path.

Instructor Answer & Exact Fix:
Check these 3 items with the attendee:
1. **Exact Path & Case:** Must be `.agents/skills/<skill-name>/SKILL.md` (`SKILL.md` in ALL CAPS, inside `.agents/skills/`).
2. **Valid YAML Frontmatter:** Line 1 of `SKILL.md` must be `---`, followed by `name:` and `description:`, closed by `---`.
3. **Trigger Keywords in Description:** Antigravity only loads `name` and `description` initially. Make sure `description` clearly states when to activate (e.g., *'Use when generating an executive SLA revenue risk audit...'*).

[Lab 02 — Rules & Skills] Q11: 'When Antigravity runs my skill script (.agents/skills/.../scripts/run_audit.py), it fails with ModuleNotFoundError: No module named src.'

Root Cause: Python does not automatically add the workspace root to `sys.path` when executing a script nested four directories deep inside `.agents/skills/<name>/scripts/`.

Instructor Answer & Exact Fix:
• Either run the script with `PYTHONPATH=. .venv/bin/python3 .agents/skills/.../scripts/run_audit.py`, OR
• Add `sys.path.insert(0, str(Path(__file__).resolve().parents[4]))` at the top of `run_audit.py` before importing from `src.*`.

[Lab 03 — FastMCP & Hooks] Q12: 'My FastMCP server works when I import it in Python, but Antigravity says MCP connection failed or invalid JSON-RPC frame!'

Root Cause: The #1 cause of `stdio` MCP bugs: someone added a `print('Connected to DB...')` statement that writes to `stdout`!

Instructor Answer & Exact Fix:
• Explain how `stdio` MCP transport works: Antigravity spawns `mcp_server/*_mcp.py` as a subprocess and communicates over `stdin` and `stdout` using strict JSON-RPC 2.0 messages.
• If any code prints plain text to `stdout`, it corrupts the JSON-RPC stream! Any debug logs inside an MCP server MUST use `print(..., file=sys.stderr)` or Python's `logging` module configured to `stderr`.
• Also verify `.agents/mcp_config.json` uses `.venv/bin/python3` (not system `python3` which might lack the `mcp` package).

[Lab 03 — FastMCP & Hooks] Q13: 'My PreToolUse hook script exits with code 1 on a violation, why didn't Antigravity treat it as a deterministic policy block?'

Root Cause: In Antigravity's hook specification, exit code `1` indicates a non-fatal script crash/warning, whereas **exit code `2`** is the explicit signal for a deterministic policy block!

Instructor Answer & Exact Fix:
• Review the Antigravity Hook Exit Code Contract:
- `sys.exit(0)` -> Action is compliant; proceed with tool execution.
- `sys.exit(2)` -> **Deterministic Policy Block!** Antigravity aborts the tool call immediately and feeds the hook's `stderr` explanation back to the agent.
• Also ensure the script reads the tool payload via `payload = json.load(sys.stdin)` and writes an audit entry to `logs/*.jsonl` before calling `sys.exit(2)`.

[Lab 04 — ADK, adk web, A2A & Terraform] Q14: 'When I launch ./scripts/run_adk_web.sh (http://localhost:8080) and send a chat message in the ADK Dev UI, it asks for GOOGLE_API_KEY or Vertex AI credentials. How does that work?'

Root Cause: `pytest` and the FastAPI A2A endpoint include a deterministic offline engine so local tests always pass without external API keys, whereas `adk web` invokes the live Gemini LLM inside `google.adk.agents.LlmAgent`.

Instructor Answer & Exact Fix:
• To chat live with Gemini inside `adk web`, configure either:
- **Option A (Vertex AI / Gemini Enterprise GCP Project):**
`export GOOGLE_GENAI_USE_VERTEXAI=TRUE`
`export GOOGLE_CLOUD_PROJECT=<your-workshop-gcp-project-id>`
`export GOOGLE_CLOUD_LOCATION=us-central1`
`gcloud auth application-default login`
- **Option B (Gemini API Key):**
`export GOOGLE_API_KEY=<your-api-key>`
• Then restart `./scripts/run_adk_web.sh` and open `http://localhost:8080`!

[Lab 04 — ADK, adk web, A2A & Terraform] Q15: 'In the ADK Dev UI (http://localhost:8080), the agent dropdown menu is empty or doesn't list telecom_nexus / retail_pulse!'

Root Cause: `adk web` expects a specific directory convention (`adk_agents/<agent_name>/__init__.py` and `agent.py` exporting a variable named `root_agent`).

Instructor Answer & Exact Fix:
• Always launch `adk web` via `./scripts/run_adk_web.sh` (which runs `.venv/bin/adk web adk_agents --port 8080` from the repository root).
• Verify that `adk_agents/<agent_name>/agent.py` defines `root_agent = LlmAgent(...)` and `adk_agents/<agent_name>/__init__.py` contains `from . import agent`.

[Lab 04 — ADK, adk web, A2A & Terraform] Q16: 'When I run ./deploy/deploy_gcp.sh to test the Terraform deployment, I don't have Docker Desktop running or GCP project owner permissions on my laptop.'

Root Cause: Attendee ran `./deploy/deploy_gcp.sh` without `--dry-run` on a laptop without Docker or active `gcloud` credentials.

Instructor Answer & Exact Fix:
• For local verification on attendee laptops, run `./deploy/deploy_gcp.sh --dry-run` (or `./labs/lab-04-multi-agent-capstone/deploy_gcp.sh --dry-run`).
• Explain that `--dry-run` validates all Terraform HCL files (`infra/terraform/main.tf`, `variables.tf`, `outputs.tf`), Dockerfile, and Gemini Enterprise registration payloads 100% locally without requiring Docker daemon or live GCP privileges!